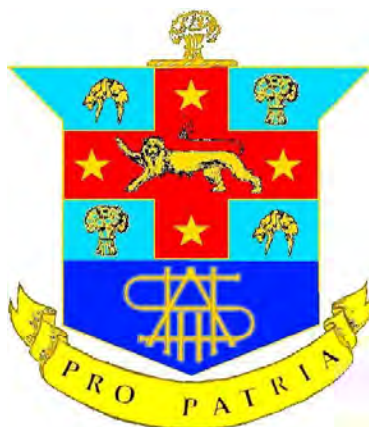




Physics

HSC Course

2010

Year 12 Trial HSC Examination

Total marks 100

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Attempt all questions
- Write using blue or black pen
- Draw diagrams using pencil
- Approved calculators may be used
- Write your I.D. number on each answer sheet
- Liquid paper must NOT be used on this paper
- For your convenience, the multiple choice answer sheet and the data sheets at the back may be removed from the rest of the paper

Total marks – 100

Section I

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

Attempt questions 1-20 (multiple choice)

Allow about 35 minutes for this part.

Part B – 55 marks

Attempt questions 20-33

Allow about 1 hour and 40 minutes for this part

Section II – Medical Physics Module

25 marks

Attempt question 34

Allow about 45 minutes for this part.

Answer this section in a separate booklet.

Teachers: Mr Coombes, Mr Robson

Task Weighting: 40 %

Multiple-choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐

(B) ☒

(C) ☐

(D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒

(B) ☒

(C) ☐

(D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word correct and drawing an arrow as follows:

correct
↓

(A) ☒ (B) ☒ (C) ☐ (D) ☐

Section I – Part A

1. Why do all objects at any particular point on the Earth's surface accelerate at the same rate in free fall?

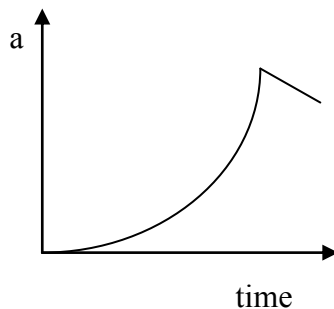
(A) The gravitational force acting on each object is the same.
(B) The gravitational force on each object is proportional to its mass
(C) The acceleration is directly proportional to the gravitational force on each mass.
(D) The acceleration is directly proportional to the mass of each object.

2. Travelling at near light speeds presents the possibility that humans may one day be able to travel to galaxies far away. Which choice correctly explains the reason for this?

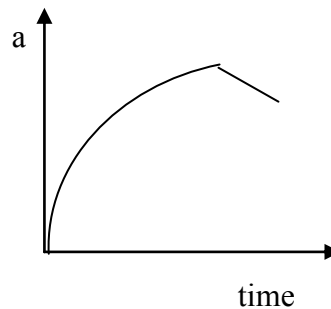
(A) Near light speeds allow humans to travel much further in the same amount of time.
(B) Time dilation allows for a person on board a space craft to age more slowly.
(C) Length contraction allows for the distance to far away galaxies to contract.
(D) Both space and time change for a person on board the space craft.

3. Which of the following acceleration versus time graphs best describes the acceleration of a rocket for the first few minutes after launch?

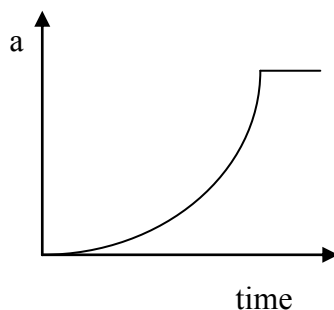
(A)



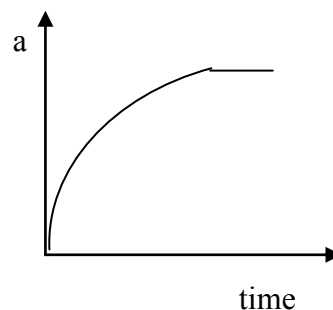
(B)



(C)



(D)



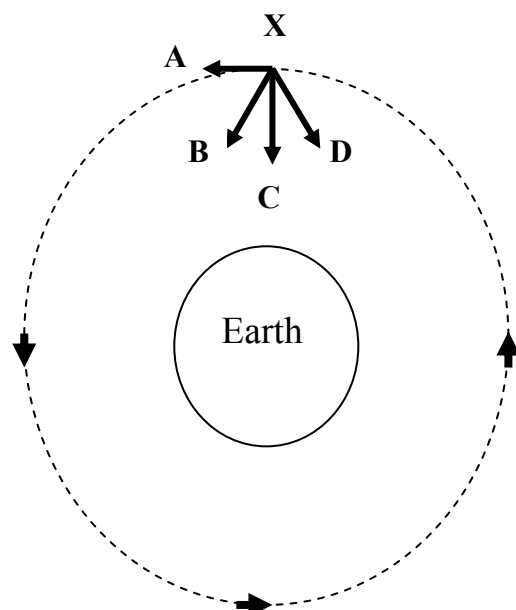
4. A new planet called Pitt was discovered in a distant galaxy. It has a radius that is 10 times that of the Earth and a mass that is 5 times that of the Earth. Which value is closest to the acceleration due to gravity on planet Pitt.

- (A) 3.92 ms^{-2}
(B) 4.90 ms^{-2}
(C) 19.6 ms^{-2}
(D) 24.5 ms^{-2}

5. Which choice correctly identifies the reason why the term g-force is used to compare the effects of launch on different astronauts.

- (A) Astronauts have different tolerances for g-force but not other forces.
(B) G-forces are much stronger than gravitational forces.
(C) Astronauts typically have different masses.
(D) Astronauts will become unconscious if the g-force exceeds 3.

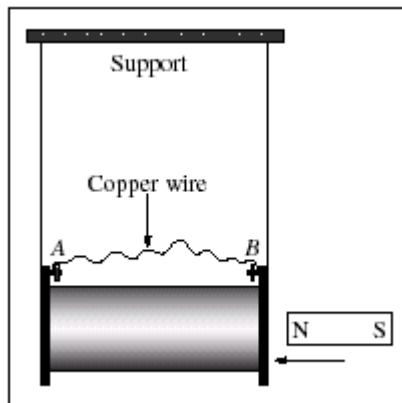
6. A low-earth orbiting satellite moves anti-clockwise around the Earth as shown below. Considering all forces acting on the satellite, which option correctly shows the direction of the net force acting on this low-earth orbiting satellite at point X.



7. Two space ships approach each other both travelling at $0.5c$ relative to an outside observer. The captain of P sends a microwave message to the captain of Q. Which choice regarding the relative speeds of the spaceships and the messages they receive is correct.

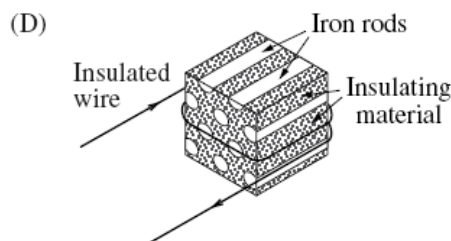
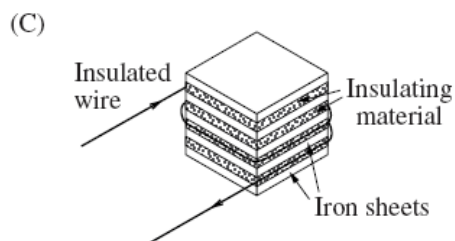
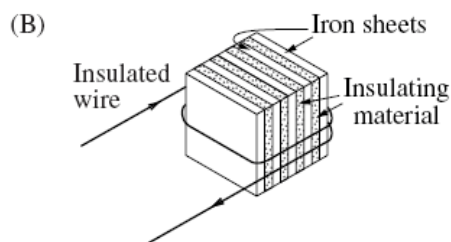
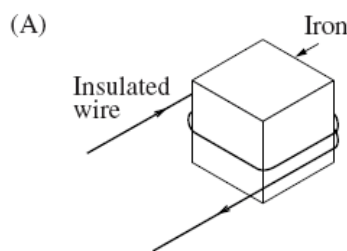
	Speed of ships relative to each other	Speed of message captain of Q.
A	c	More than c .
B	c	c
C	Less than c	More than c
D	Less than c	c

8. A student conducted an investigation by pushing a magnet into a solenoid. The solenoid was suspended so that it was free to move.

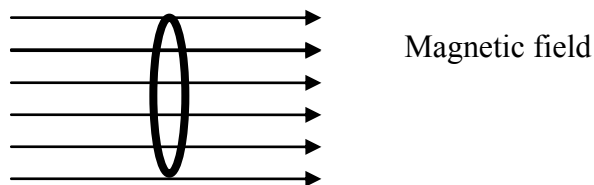


Which statement is the best explanation of what would happen in this investigation?

- (A) The solenoid will move towards the magnet due to the induced current in the solenoid.
(B) The solenoid will move away from the magnet due to the induced current in the solenoid.
(C) The solenoid will not move because there is no change in flux through the solenoid.
(D) The solenoid will not move because solenoid has no power supply.
9. Which of the following designs would produce the most efficient transformer?



10. A metal ring has its plane perpendicular to a magnetic field as shown.

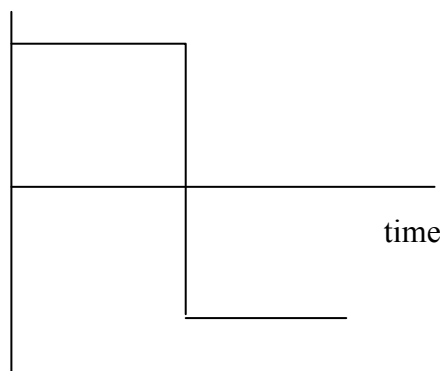


Over a period of time, the strength of the magnetic field is reduced at a uniform rate from 0.5T to zero. Then its direction is reversed and the strength is increased, at the same uniform rate back up to 0.5T.

Which graph shows a possible emf induced in the ring during this time?

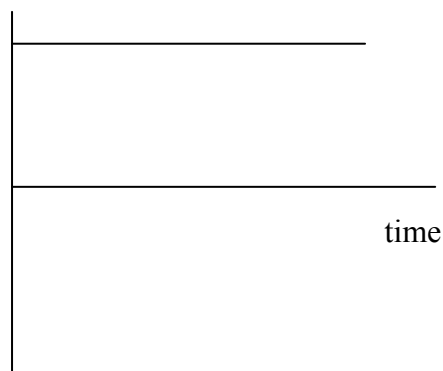
(A)

emf



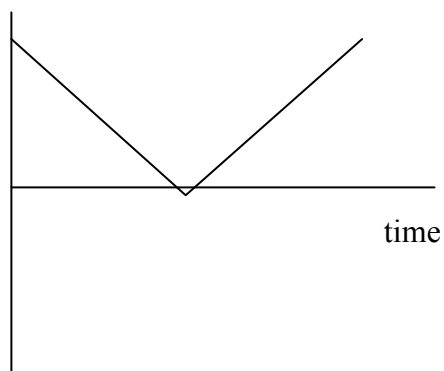
(B)

emf



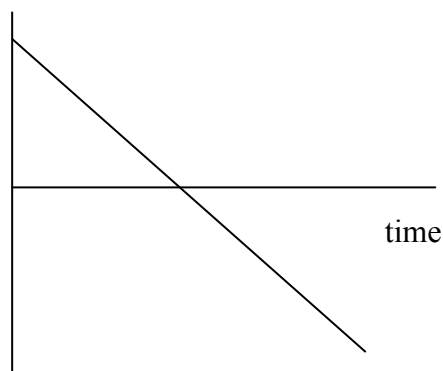
(C)

emf

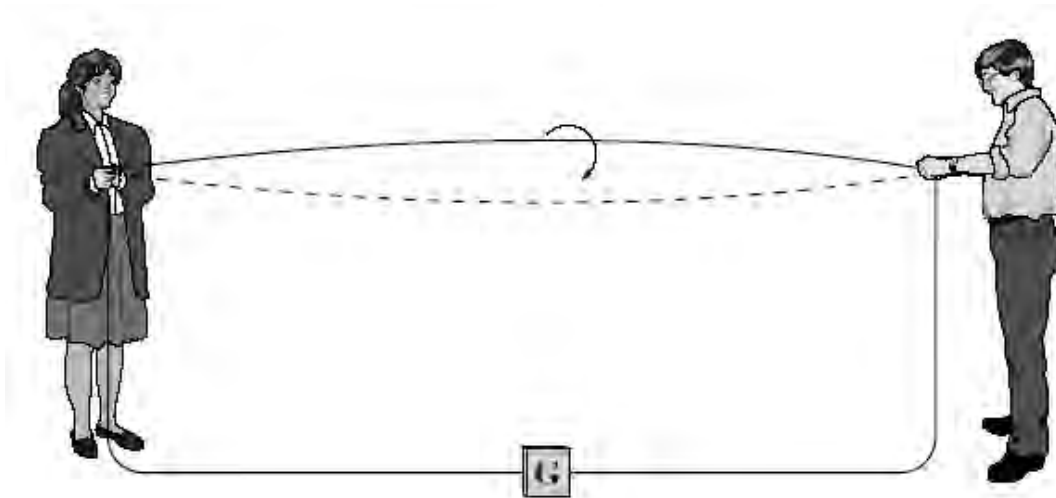


(D)

emf



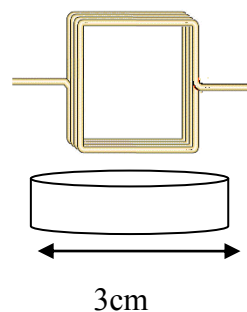
11. Two students carried out an investigation to measure the induced current by moving a wire in the Earth's magnetic field.



Which choice shows how the investigation should be carried out to maximise the current induced in the wire. Assume that an equal force is applied to the wire regardless of the type of motion used.

	Orientation of wire	Motion of wire
A	East-West	Up and down
B	East-West	Circular motion
C	North-South	Up and down
D	North-South	Circular motion

12. A student designs an electric motor by placing a square loop of wire above a 3cm diameter neodymium magnet as shown below.



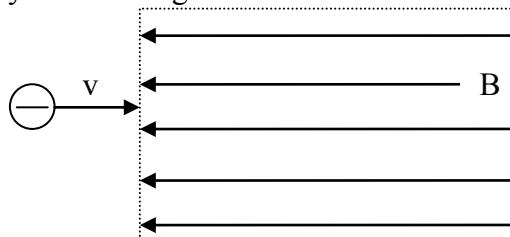
If the student makes the loop out of a 1.2m long piece of wire, how many square loops should the student wind in order to produce the maximum amount of torque?

- (A) 5
- (B) 10
- (C) 15
- (D) 20

13. Magnetic flux density is :

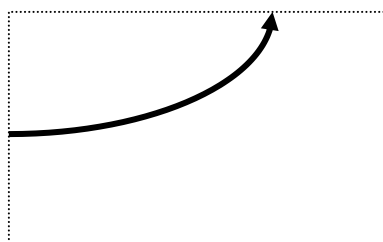
- (A) the same as magnetic field strength
- (B) the magnetic field strength divided by the area through which it passes.
- (C) proportional to the amount of flux lines through a loop of any size.
- (D) equivalent to the force acting on an electron within the field.

14. A negatively charged particle enters a region with a uniform magnetic field. The direction of the particle's velocity and the magnetic field are shown in the diagram

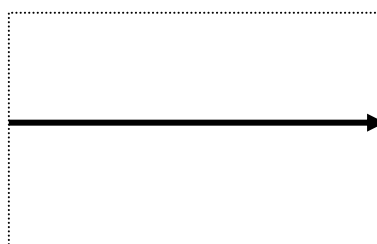


Which diagram best represents the subsequent path of the particle in the magnetic field?

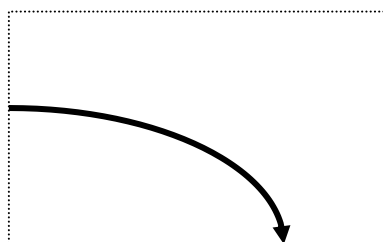
(A)



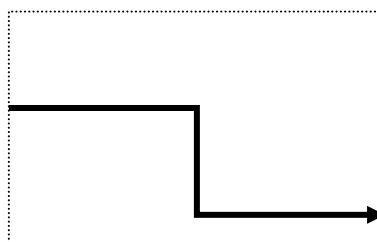
(B)



(C)



(D)



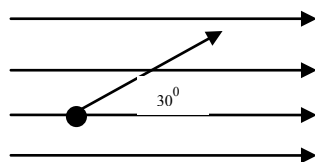
15. A light ray of wavelength λ is incident on a metallic surface in a vacuum. No electrons are emitted from the surface.
Which of the following is most likely to cause electrons to be emitted from the surface of the metal?

- (A) Decrease the frequency of the incident light.
- (B) Decrease the wavelength of the incident light.
- (C) Increase the intensity of the incident light.
- (D) Use a metal with a greater work function.

16. A photon of light has an energy of 3.75×10^{-19} joules. What is its wavelength?

- (A) 450 nm
- (B) 490 nm
- (C) 530 nm
- (D) 560 nm

17. A proton enters a magnetic field at an angle of 30° as shown in the diagram below. The proton has a velocity of $3.0 \times 10^6 \text{ m s}^{-1}$ and experiences a force of $1.2 \times 10^{-13} \text{ N}$ into the page.

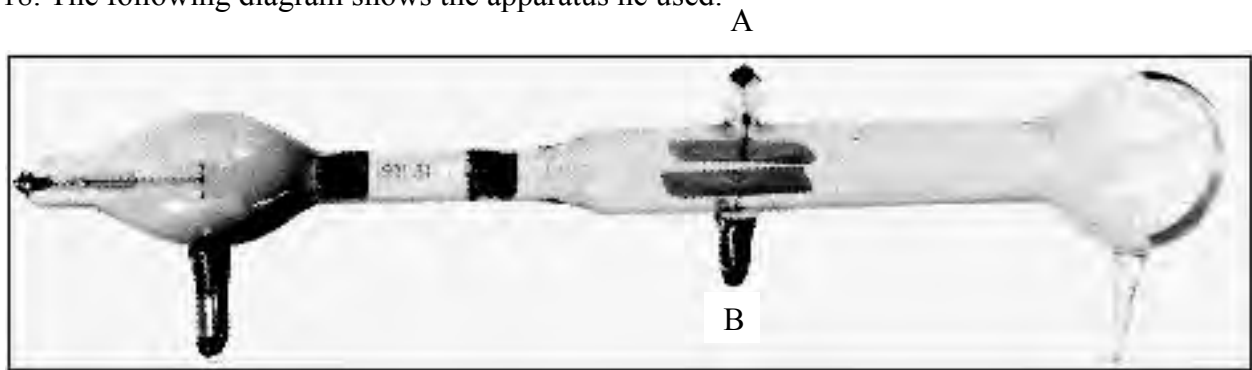


The magnitude of the magnetic flux density is

- (A) 0.25 T
- (B) 0.5 T
- (C) 1.25 T
- (D) 5.0 T

J.J. Thomson carried out an experiment to determine the charge to mass ratio of electrons.

18. The following diagram shows the apparatus he used.



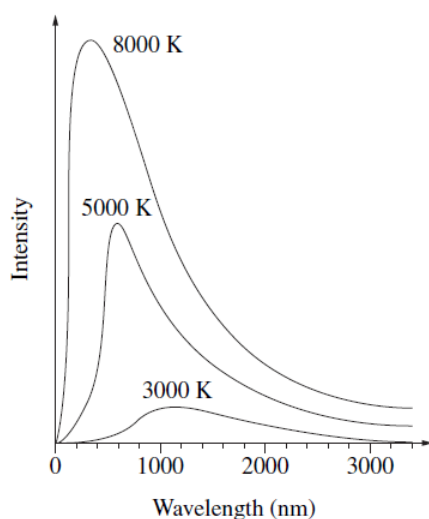
An electric field is produced by applying a voltage across plates A-B, with B being positive. Which of the following choices best represents the direction of a magnetic field which, when applied to the region of the charged plates, would cause the cathode ray to pass through the tube undeflected?

- (A) vertically down the page
- (B) vertically up the page
- (C) into the page
- (D) out of the page

19. Between 1886 and 1889, Heinrich Hertz performed a series of experiments which determined the speed of the newly discovered radio waves. This involved measuring their wavelength. What method was used to do this?

- (A) The radio waves were passed through a glass sheet and the effect on the spark detector observed.
- (B) The radio wave beam was split into two using a half-silvered mirror and recombined.
- (C) The effect of the radio waves on cathode rays in a cathode ray tube was observed.
- (D) The interference pattern produced when radio waves combined was observed.

20. The graph shows the intensity-wavelength curves for blackbodies at different temperatures. Similar results obtained by experimental physicists at the end of the 1800s were a major problem because they could not be explained using classical wave physics.



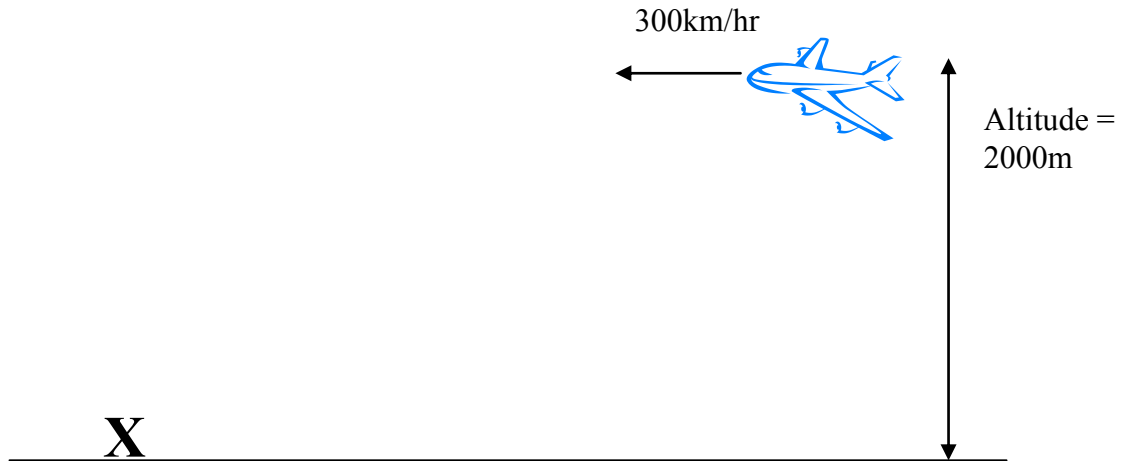
In 1900, Max Planck proposed a mathematical model which produced theoretical results which matched the experimental ones.

What assumption was his model based on?

- (A) Light is quantised and its frequency is directly proportional to its intensity.
- (B) Light is a wave with a frequency that is directly proportional to its energy.
- (C) Light is quantised and its frequency is directly proportional to the energy of the quanta.
- (D) Light is a wave with a frequency that is inversely proportional to its wavelength

Section I – Part B

21. A bomber travels at 300 km/hr above its desired target as shown below. At what distance before the target does the bomb have to be released in order to hit the target. Ignore air resistance in your calculations. (**3 marks**)



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22. Saturn has a moon called Tethys which has an orbital period of 1.9 days and an orbital radius of 295,000km. Calculate the mass of Saturn. (**3 marks**)

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23. Recount the method used in the Michelson-Morley experiment and describe the result that they anticipated. **(4 marks)**

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24. An observer on Earth is watching a spaceship of rest length 50m moving away from Earth at $0.6c$.

(a) Calculate the length of the ship from the observer's point of reference. **(2 marks)**

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(b) The observer falls asleep when the clock on board the spaceship and his watch both read exactly 10am. When he wakes up the times differ by 10 minutes.

What time is it on the observer's watch? **(3 marks)**

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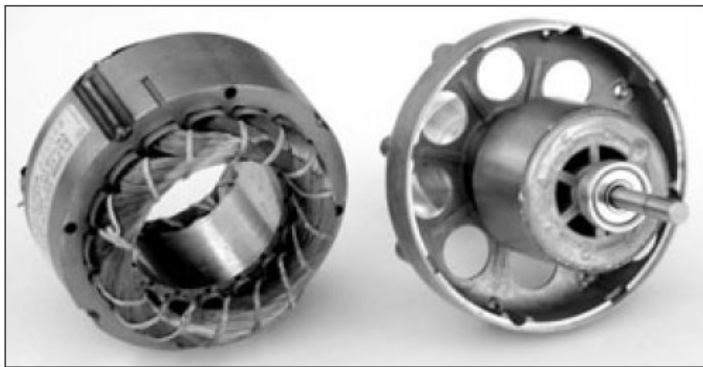
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25. The following picture shows a commonly used electric motor.



Compare the construction of this motor with a simple DC electric motor. **(3 marks)**

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26. Assess the importance of transformers in providing a safe and efficient supply of electrical energy to homes. **(6marks)**

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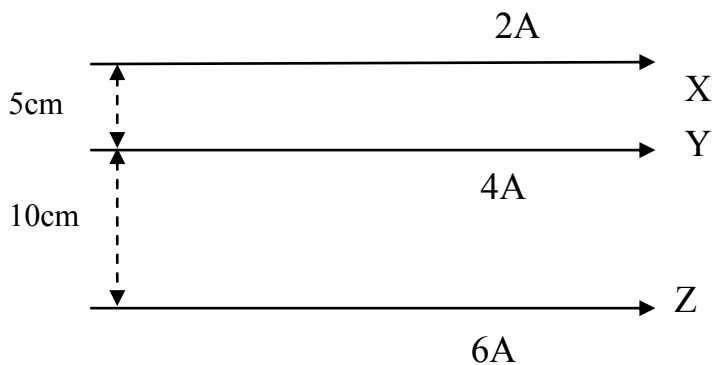
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27. Three long, straight, parallel wires; X, Y and Z are 5cm and 10cm apart respectively and carry currents of 2A, 4A and 6A. The currents in wires X and Y are in the opposite direction to the current in wire Z.



Calculate the magnitude AND direction of the force per unit length on wire Y, due to the other two wires. **(3 marks)**

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28. The motion of the Earth and other planets is utilized when sending space probes to planets further from the sun in the solar system. Explain why this is the case. **(4 marks)**

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29. Explain why the wires of the secondary coil in a step up transformer are a different thickness to the wires in the primary coil. **(3 marks)**

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30. Recount the experiment that you performed in class to measure the effect on a generated electric current when the distance between the coil and magnet is varied. **(4 marks)**

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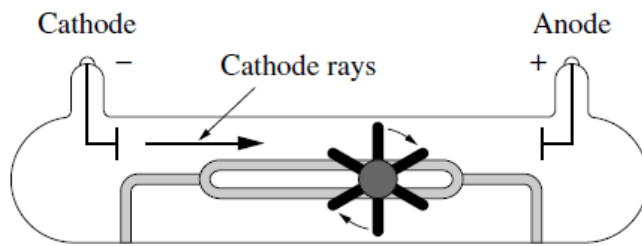
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31. During the nineteenth century, scientists performed a variety of experiments using cathode ray tubes. One tube contained a glass wheel as shown below.



(a) Outline one observation about cathode rays that was made using this type of tube. **(1 mark)**

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(b) Assess the importance of this observation in determining whether cathode rays were waves or charged particles. **(3 marks)**

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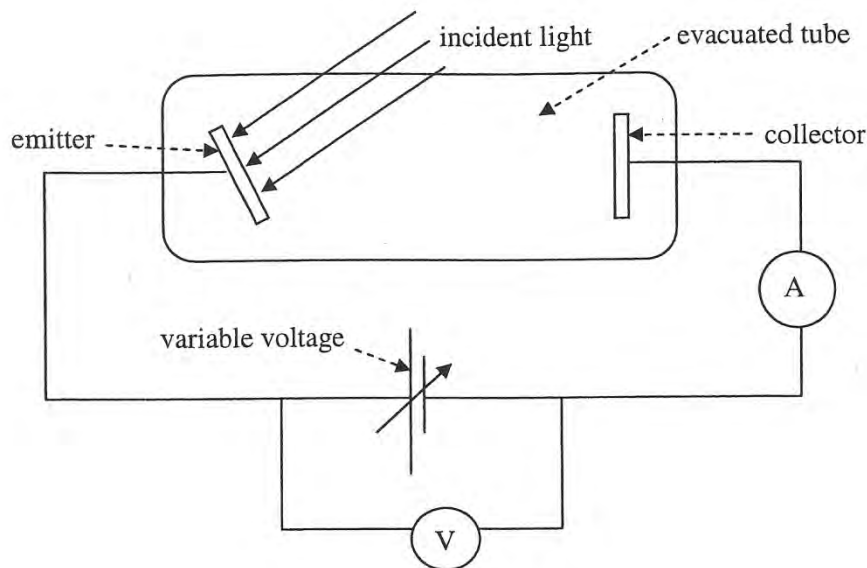
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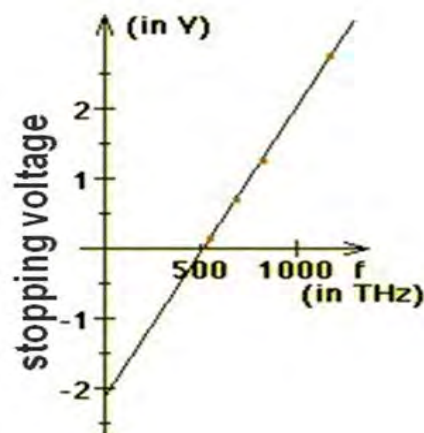
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32. The diagram shows equipment set up to investigate the photoelectric effect. Light of different frequencies is shone onto the emitter and the voltage adjusted to just prevent electrons reaching the collector. In this situation, the work done by the applied electric field reduces the kinetic energy of the photoelectrons to zero.



The results of one experiment using a particular metal emitter are shown in the graph below.



(Note: a THz = 1×10^{12} Hz)

(a) Identify the threshold frequency for the metal emitter. (1 mark)

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(b) The work function of a metal can be calculated using the following formula

$$\text{work function} = \text{energy in a photon} - \text{kinetic energy of photoelectron}$$

Using the graph, calculate the work function of the metal emitter.(2 marks)

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(b) Assess the impact of each advance identified in part (a) on the development of the television.
(4 marks)

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End of Part B

Section II
Option – Medical Physics – 25 marks

Question 34: Answer this question in a separate booklet

a) (i) Define the term acoustic impedance.

(1 mark)

(ii) The table provides information about the acoustic properties of some body tissues

Body Tissue	Density (kg m^{-3})	Velocity (m s^{-1})
Blood	1060	1570
Brain	1025	1540
Fat	925	1450
Muscle	1075	1590

Calculate the acoustic impedance of brain tissue.

(2 marks)

(iii) A stroke, or cerebrovascular accident, occurs when blood supply to part of the brain is disrupted, causing brain cells to die. **Using information in the table** and an appropriate calculation, determine whether an ultrasound image of the brain is likely to successfully identify the location of a damaged blood vessel.

(3 marks)

(b) Below is an X-ray image and a CAT scan of the same part of the body of a patient suffering an unusual medical condition.



X-ray image



CAT scan

(i) Explain how the CAT scan was produced.

(3 marks)

(ii) “CAT scans are always a better diagnostic tool than conventional X-ray images”. Assess this statement with reference to this patient’s medical condition.

(5 marks)

(c) The next question refers to the table below

Isotope of Iodine	Half Life (days)	Decay Products
Iodine-123	0.55	gamma rays
Iodine-125	59	gamma rays
Iodine-131	8.00	beta particles, gamma rays

(i) Calculate the amount of iodine-131 remaining after 24 hours if the initial mass was 200 mg.

(1 mark)

(ii) Identify which isotope listed above would be most suitable to image the thyroid gland. Justify your answer with reference to information in the table. **(3 marks)**

(iii) Describe how the radioactive isotope of iodine can be used to image the thyroid gland. **(3 marks)**

(d) Analyse the relationship between scientific advances and medical imaging techniques. **(5 marks)**

END OF EXAM

HIGHER SCHOOL CERTIFICATE EXAMINATION

Physics

DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2} a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$v_{av} = \frac{\Delta r}{\Delta t}$$

$$a_{av} = \frac{\Delta v}{\Delta t} \text{ therefore } a_{av} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2} mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_f}{R_i}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

KEY		Atomic Number		Symbol of element		Name of element	
79	Au	197.0	Gold	5	B	10.81	Boron
26	Fe	55.85	Iron	13	Al	26.98	Aluminium
44	Ru	101.1	Ruthenium	31	Ga	69.72	Gallium
43	Tc	[97.91]	Technetium	30	Zn	65.41	Zinc
75	Re	186.2	Rhenium	48	Cd	112.4	Cadmium
76	Os	190.2	Osmium	47	Ag	107.9	Silver
108	Hs	[277]	Hassium	46	Pd	106.4	Palladium
107	Bh	[264]	Bohrium	45	Rh	102.9	Rhodium
106	Sg	[266]	Seaborgium	77	Ir	192.2	Iridium
105	Db	[262]	Dubnium	109	Mt	[268]	Meitnerium
104	Rf	[261]	Rutherfordium	110	Ds	[271]	Darmstadtium
88	Ra	[226]	Radium	111	Rg	[272]	Roentgenium
87	Fr	[223]	Francium				
132.9	Cs		Cesium	81	Tl	204.4	Thallium
85.47	Rb		Rubidium	80	Hg	200.6	Mercury
56	Ba		Barium	79	Au	197.0	Gold
57-71			Lanthanoids	78	Pt	195.1	Platinum
72	Hf		Hafnium	77	Ir	192.2	Iridium
73	Ta		Tantalum	76	Os	190.2	Osmium
74	W		Tungsten	75	Re	186.2	Rhenium
74	W		Tungsten	74	W	183.8	Tungsten
73	Ta		Tantalum	73	Ta	180.9	Tantalum
72	Hf		Hafnium	72	Hf	178.5	Hafnium
57-71			Lanthanoids	40	Zr	91.22	Zirconium
38	Sr		Strontium	39	Y	88.91	Yttrium
37	Rb		Rubidium	41	Nb	92.91	Niobium
36	Kr		Krypton	42	Mo	95.94	Molybdenum
35	Br		Bromine	43	Tc	[97.91]	Technetium
34	Se		Selenium	44	Ru	101.1	Ruthenium
33	As		Arsenic	45	Rh	102.9	Rhodium
32	Ge		Germanium	46	Pd	106.4	Palladium
31	Ga		Gallium	47	Ag	107.9	Silver
30	Zn		Zinc	48	Cd	112.4	Cadmium
29	Cu		Copper	49	In	114.8	Indium
28	Ni		Nickel	50	Sn	118.7	Tin
27	Co		Cobalt	51	Sb	121.8	Antimony
26	Fe		Iron	52	Te	127.6	Tellurium
25	Mn		Manganese	53	I	126.9	Iodine
24	Cr		Chromium	54	Xe	131.3	Xenon
23	V		Vanadium	55	Cs		Cesium
22	Ti		Titanium	56	Ba		Barium
21	Sc		Scandium	57-71			Lanthanoids
20	Ca		Calcium	89-103			Actinoids
19	K		Potassium	88	Ra		Radium
12	Mg		Magnesium	87	Fr		Francium
11	Na		Sodium	86	Rn		Radon
10	Ne		Neon	85	At		Astatine
9	F		Fluorine	84	Po		Polonium
8	O		Oxygen	83	Bi		Bismuth
7	N		Nitrogen	82	Pb		Lead
6	C		Carbon	81	Tl		Thallium
5	B		Boron	80	Hg		Mercury
4	Be		Beryllium	79	Au		Gold
3	Li		Lithium	78	Pt		Platinum
2	He		Helium	77	Ir		Iridium
1	H		Hydrogen	76	Os		Osmium

Lanthanoids

57	La	138.9	Lanthanum	65	Tb	158.9	Terbium	66	Dy	162.5	Dysprosium	67	Ho	164.9	Holmium	68	Er	167.3	Erbium	69	Tm	168.9	Thulium	70	Yb	173.0	Ytterbium	71	Lu	175.0	Lutetium
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Actinoids

89	Ac	[227]	Actinium	93	Np	[237]	Neptunium	94	Pu	[244]	Plutonium	95	Am	[243]	Americium	96	Cm	[247]	Curium	97	Bk	[247]	Berkelium	98	Cf	[251]	Californium	99	Es	[252]	Einsteinium	100	Fm	[257]	Fermium	101	Md	[258]	Mendelevium	102	No	[259]	Nobelium	103	Lr	[262]	Lawrencium
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For elements that have no stable or long-lived nuclides, the mass number of the nuclide with the longest confirmed half-life is listed between square brackets.

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (October 2005 version) is the principal source of data. Some data may have been modified.

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Multiple-choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐ (B) ☒ (C) ☐ (D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word correct and drawing an arrow as follows:

correct
↓

(A) ☒ (B) ☒ (C) ☐ (D) ☐

Part A

- | | | | | |
|-----|---------------------------|---------------------------|---------------------------|---------------------------|
| 1. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 2. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 3. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 4. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 5. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 6. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 7. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 8. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 9. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 10. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 11. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 12. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 13. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 14. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 15. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 16. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 17. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 18. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 19. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |
| 20. | (A) ☐ | (B) ☐ | (C) ☐ | (D) ☐ |

Marking Criteria

M/C: 1 B 2 C 3 A 4(na) 5C 6D 7D 8B 9B 10B 11A 12B 13A 14B 15B 16C 17B 18D 19D 20C

21. Criteria	Mark/s
Uses and appropriate method to calculate a distance of 1684m	3
Correctly calculates the time of fall OR Uses the correct method but with one incorrect substitution.	2
Uses the correct method but with two errors.	1

22. Criteria	Mark/s
Uses an appropriate method to calculate a mass of 5.6×10^{26} kg	3
Correctly method of calculation in kg but with one incorrect substitution OR Correct answer with incorrect units.	2
Correctly method of calculation but with two incorrect substitutions.	1

23. Criteria	Mark/s
Outlines the method including a diagram or detailed description of the apparatus and the rotation of the apparatus AND Describes the expected result which is a change in the interference pattern after rotation.	4
Method correct without a description of the expected result or one aspect of method omitted.	3
Two important aspects omitted.	2
Three important aspects omitted.	1

24a. Criteria	Mark/s
Correct calculation of answer of 40m.	2
Correct method with one error(not including calculation of longer length)	1

24b. Criteria	Mark/s
Correct calculation of time difference and identifies time on observers watch as 10:50am.	3
Correct calculation of time difference but incorrect time on observers watch..	2
Identifies that the observer's time is equal to the time on board plus 10 minutes.	1

25. Criteria	Mark/s
Three significant differences or two differences and one similarity using comparative sentences.	3
Two significant similarities or differences with no incorrect comparisons. Must use comparative sentences	2
One significant similarity or difference with no incorrect comparisons.	1

26. Criteria	Mark/s
Judgement of importance AND Significant reasoning related to safety and efficiency AND Demonstrates a logical progression and good use of scientific terminology	6
Judgement of importance AND Significant reasoning related to safety and efficiency	4-5
Limited discussion of issues related to safety and efficiency	2-3
One correct point related to efficiency or safety.	1

27. Criteria	Mark/s
Answer calculates the force per unit length due to wire X and Y and adds them using appropriate vector addition.	3
Answer calculates the force per unit length due to wire X and Y and adds them incorrectly. OR Correct method with one error or substitution..	2
Correct method with two errors	1

28. Criteria	Mark/s
Answer explains how the Earth's orbital motion and the slingshot effect can be utilised to minimise the fuel needed by	4

Criteria	Mark/s
the probe.	
Answer demonstrates some understanding of how orbital motion of the Earth and the slingshot effect are utilised.	2-3
Answer identifies that the slingshot effect or orbital motion of the Earth is utilised.	1

29. Criteria	Mark/s
Explanation of why increasing voltage decreases current and relates the change in current to the wires in the secondary being thinner.	3
Explanation of why increasing voltage decreases current.	2
Identifies that increasing voltage decreases current.	1

30. Criteria	Mark/s
Correct description of method including - A suitable method of controlling speed. - The use of a galvanometer or micro ammeter - Measurements across a specific range of distances. - Repetition of each measurement at least three times.	4
Correct description of method with one of the above omitted.	3
Correct description of method with two of the above omitted.	2
Correct description of method with three of the above omitted.	1

31a. Criteria	Mark/s
Clearly outlines one OBSERVATION (eg the paddle wheel rotates when the beam is turned on) – no marks if an inference is included!	1

31b. Criteria	Mark/s
States that the observation is VERY/QUITE important and Clearly and correctly links the observation to a particle property (eg rotating wheel implies cathode rays have momentum and therefore mass) and States that this is a property of particles not waves	3
Links the observation to a particle property but does not assess the importance OR Clearly assesses the importance but does not establish a link between the observation and the particle property	2
Response contains a link between the observation and a relevant particle property	1

32a. Criteria	Mark/s
$F = 500 \text{ THz}$ or $5.00 \times 10^{14} \text{ Hz}$ (must have correct units)	1

32b. Criteria	Mark/s
Calculation clearly shows that the work function equals the photon energy when the stopping voltage (and hence E_k) is zero (x-intercept on graph gives threshold frequency). Work function = $3.313 \times 10^{-19} \text{ J}$	2
Calculates $E = hf$ as $3.313 \times 10^{-19} \text{ J}$	1

32c. Criteria	Mark/s
A well-structured response which states that the experiments were VERY significant (evaluate statement) and - outlines the failure of classical wave physics to explain the results of experiments on the photoelectric effect and blackbody radiation and that therefore - there was a requirement for a new model of light – the particle model which was - the start of a new way of scientific thinking – quantum physics	3-4
Response outlines some relevant experiments and their	1-2

results but does not clearly link this to failure of classical theory and hence the requirement of a new model of light.	
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33a. Criteria	Mark/s
Outlines a scientific advance related to each device (eg the discovery that an electric current produces a magnetic field leads to solenoids being used to produce magnetic fields in a TV; the discovery that cathode rays were a stream of negatively charged particles leads to CR tubes being used in a TV)	2
Outlines one advance that relates to one device	1

33b. Criteria	Mark/s
Response contains a clear assessment of the impact (eg the impact was great, very significant etc) and Clearly links each advance outlined in part (a) to the use of each device in a TV	3-4
Response clearly links each advance outlined in part (a) to the use of each device in a TV	3
Response establishes a clear link between one advance and the use of a technology in the TV	2
Response outlines the role of each device in the TV	

34 a (i). Criteria	Mark/s
Defines acoustic impedance as the product of a material's density and the acoustic velocity ($Z = \rho v$)	1

34 a (ii). Criteria	Mark/s
Correctly calculates the acoustic impedance as $1578500 \text{ kgm}^{-2}\text{s}^{-1}$ or rayls (must have correct units)	2
Calculates correct value without units	1

34 a (iii) Criteria	Mark/s
Response states that an ultrasound image would NOT successfully locate the blood vessel AND Justifies this by correctly calculating the percentage of US reflected from a brain/blood boundary (0.07 %) and that therefore the reflected signal would have too low an intensity to be detected	3
States no and performs the relevant calculation but does not link this clearly to the detection of the boundary.	2
Performs the relevant calculation correctly	1

34 b (i). Criteria	Mark/s
A thorough description which includes - a fan-shaped beam of X-rays is passed through the body - The X-ray tube is rotated 180° around the body - a ring of detectors on the opposite side of the body detects variations in the intensity of transmitted X-rays - data analysed by computer (A diagram was useful to illustrate these features)	3
Response contains the above points but lacks detail and/or clarity (eg no diagram)	2
Response missing one essential point	1

34 b (ii). Criteria	Mark/s
Response includes a sentence stating that the statement is incorrect an indication that the nail can only be seen in the X-ray and not the CAT scan a comprehensive discussion of the advantages/disadvantages of CAT scans as opposed to X-rays (at least 3) that are relevant to the patient in the question	5
A comprehensive response but No clear assessment made OR No indication that the CAT scan does not show the nail	4
Response includes a comprehensive discussion of the advantages/disadvantages of CAT scans as opposed to X-rays (at least 3) but does not relate it to the patient	3
A general and superficial discussion of the uses of X-rays and	1-2

CAT scans.	
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34 c (i) Criteria	Mark/s
Response is a calculation showing an understanding of an exponential decay process	1

34 c (ii) Criteria	Mark/s
Identifies I-123 and states that its shorter half life means that the patient will not be exposed to gamma radiation for too long (potential to ionise body chemicals or cause radiation poisoning) AND That the half life is still long enough for the iodine to accumulate and be metabolised in the thyroid AND That the only product is gamma rays – no highly ionising beta particles as produced by I-131	3
Response identifies I-123 but omits one of the above reasons	2
Response identifies I-123 OR Response contains the above information but includes significant errors (eg that a beta particle is involved in producing a PET scan)	1

34 c (iii). Criteria	Mark/s
A comprehensive response which includes Radioisotope is injected/ingested/inhaled and travels in blood to the thyroid Radioisotope accumulates and is metabolised in the thyroid Decay of isotope produces gamma rays which pass through the body and are detected by gamma ray cameras The data is then processed by a computer	3
Response missing one of the above or containing a significant error	2
Response missing two of the above or containing two significant errors	1

34 d Criteria	Mark/s
A well-structured response which identifies at least 2 significant scientific advances (eg induction, piezoelectric effect, radioactive decay) and clearly relates the implications of these advances to at least 3 different imaging techniques	4-5
Response identifies a single advance but also includes at two or more relevant advances in technology which are related to imaging techniques OR Response identifies relevant advances but does not clearly link them to the imaging techniques	2-3
Identifies one advance in science	1